FEWS NET
FAMINE EARLY WARNING SYSTEMS NETWORK

UKRAINE Targeted Analysis — February 18, 2022

This report on Ukraine is a targeted analysis conducted by FEWS NET's Early Warning Team in response to ongoing events in the region. FEWS NET does not have a presence in Ukraine and does not monitor Ukraine through the standard mechanisms used for monitoring and projecting food insecurity in FEWS NET's reporting countries. As such, analyses on Ukraine are based on available secondary data. Updated analyses, as available, can be found here. The analysis in this report reflects a scenario of a moderate-to-major invasion of Ukraine by Russia. Unlike with standard FEWS NET reporting, the scenario presented in this targeted analysis was not necessarily the most likely scenario.

Key Messages

In the event of a "moderate-to-major" Russian invasion into Ukraine via the Dnipro River strategy (see full scenario and map of invasion routes detailed on p. 9):

Figure 1. Map of Ukraine

Source: *Encyclopedia Britannica*

- In the near term, the direct impacts of conflict in affected areas of Ukraine east of the Dnipro[1] River would likely:
  - Significantly disrupt livelihoods, including during the agricultural growing season, through physical access constraints and damage to homes, productive assets, agricultural land, roads, and other civilian infrastructure. Millions of people would also be at risk of losing access to critical income sources, including government-provided pensions and social safety nets, should territory change hands.
  - Result in high levels of population displacement as households in affected areas east of the Dnipro River flee to safer areas, especially to western Ukraine. Hundreds of thousands to millions of displaced households separated from assets and typical sources of income would likely need immediate assistance, particularly during the ongoing winter season when fuel needs are highest. Households without sufficient resources living in temporary locations would be expected to need sustained humanitarian assistance until livelihoods are re-built in new, permanent locations.
  - Severely disrupt production of cereals (mostly wheat and maize) and other key crops including oilseeds, as areas east of the Dnipro River produce around half of national wheat, maize, and sunflower seed production. This would have a significant impact on domestic supply in eastern Ukraine.
  - Disrupt market access for traders and households, with periods of localized supply shortages possible in areas where conflict blocks transportation routes and where local production is insufficient. Prices would likely increase in the short term where supply is disrupted. It is possible that Russian forces cut off supplies of food and essential non-food commodities—such as fuel and medicine—from particular areas for prolonged periods as a strategy to pressure Ukrainian forces and authorities.
- In addition to short-term disruptions to domestic agricultural production due to the direct impacts of conflict, Ukraine's production and export potential would likely significantly reduce in the medium and long term due to challenges in accessing agricultural inputs, disruptions to port access, and trade restrictions and international sanctions imposed on Russia. This would negatively impact Ukraine's national economy due to loss of important export revenue and foreign exchange. While some producers of exportable agricultural commodities in western Ukraine could benefit from rising global prices and increased demand, benefits would be limited by the capacity of storage and port infrastructure.
- A significant and prolonged impact on international grain markets and prices would be likely due to declining Ukrainian exports to the world market. According to USDA forecasts, in the ongoing 2021/22 marketing year, Ukraine is projected to be responsible for 12 percent of global wheat exports and 17 percent of maize exports. Given generally favorable global production forecasts, many buyers would likely source wheat and maize, or cheaper substitutes, from alternative markets. However, given the high share of Ukrainian grains in total global exports, a significant shock to Ukrainian grain

[1] Other spellings include "Dnieper" and "Dnepr."

exports would reduce supply relative to earlier forecasts, driving up global prices that are already rising due to the political tensions in the region. It additionally remains possible that global production forecasts continue to be revised downward, and any impacts of international sanctions on Russia that would disrupt Russian energy and grain exports would result in even higher global prices.

  - As has been seen in previous global food price crises, poor countries that are highly reliant on imports for their staple food supplies are more vulnerable to global price shocks. Higher staple food prices in vulnerable countries would further constrain resources of poor households, exacerbating the scale and severity of existing food insecurity, with poor urban households among the worst affected due to high market dependence.
  - Several countries in the Middle East, North Africa, and Asia regions (including Lebanon, Tunisia, Libya, Pakistan, and Indonesia) also have relatively higher dependence on Ukrainian wheat and would need to source grain from other countries at higher prices. The degree of concern for their ability to do this depends on a complex set of factors beyond the level of dependence on Ukrainian grain, overall import dependence, and overall financial status, including: the structure of importation supply chains including the number and nimbleness of major traders (many of whom are private companies), the country's geographic location including proximity to alternative source markets, the amount of staple food commodities grown domestically, the strength of national institutions and ability of the government to respond with appropriate trade and social safety net policies, the degree of international and regional cooperation in response, and more.

- Given Ukraine's bumper harvest in 2021, many poor rural households likely have above-average levels of food stocks and/or financial resources. This is likely to provide some ability to cope with temporary disruptions to income-earning. However, resources among poor and smallholder households in rural areas east of the Dnipro River would be stretched thin as time goes on in the cold winter and lean months, with Stressed (IPC Phase 2)[2] outcomes likely to persist until food from the next harvest becomes available around June/July. During this time, worst-affected rural households without sufficient resources would likely experience Crisis (IPC Phase 3) outcomes in the short term as stocks are exhausted during the winter months and amidst significant disruptions to livelihoods.
- Highly populated urban areas would be devastated by the direct impacts of active conflict. Civilian casualties, high levels of displacement, and significant disruptions to livelihoods, income-earning, and markets would be likely. As soon as any invasion begins—or as the threat of invasion becomes more imminent—urban households who can afford to do so would likely stockpile food, driving up prices.[3] Poor households who cannot afford to stockpile food would likely face consumption gaps and Crisis (IPC Phase 3) outcomes during any periods of time where active conflict prevents households from accessing markets for several days or more.
- Though significant uncertainty exists, an estimated 2.5-4.99 million people (around 5 to 10 percent of the national population)—mostly displaced households—would likely need humanitarian assistance in the event of an invasion of this scale. In both rural and urban areas, should food supply be cut off for prolonged periods, Emergency (IPC Phase 4) outcomes are possible and would last until supply is restored.
- The impacts of conflict on livelihoods and business activity—including via disruptions to international exports—would be highly detrimental to Ukraine's economy, even in the long term. Russia's long-term objective could be to render Ukraine an economically inviable state. For millions of people, livelihoods would be damaged and unemployment would increase.
- While the importance of the Donbas region as a supplier of agricultural commodities to the rest of Ukraine and to the global export market has diminished since the onset of conflict in 2014, the importance of agricultural production in the region in supplying local populations has increased. It is likely that territory east of the Dnipro River would experience similar shifts. Though increased local supply due to reduced exports would likely increase availability of food for farmers' own consumption and put downward pressure on prices of locally produced commodities, rising production costs are likely to place opposing upward pressure on prices of locally produced commodities. Prices of imported goods would also likely increase. Overall, reductions in income-earning could outweigh any benefits of increased availability of food for own consumption. Additionally, in the near term, local agricultural production would also be threatened.
- Though not part of the scenario analyzed, it is possible that Russia would additionally engage in a full-scale naval blockade of the Black Sea, where it has been pursuing increased control over the years. Should this occur, both domestic and international impacts resulting from reduced exports would be significantly exacerbated.

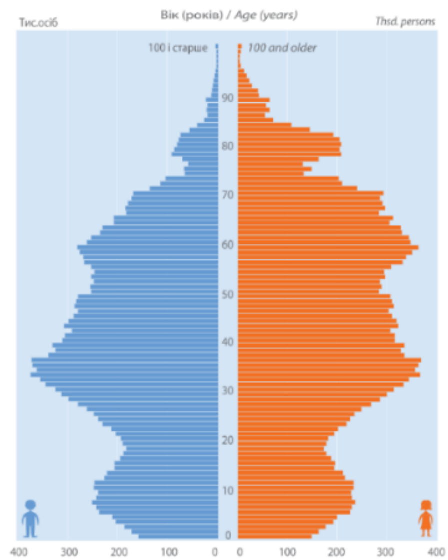
[2] The IPC is a five-phase scale, which is comparable across countries and time. The phases range from None/Minimal (IPC Phase 1) to Catastrophe/Famine (IPC Phase 5). When a household is in Crisis (IPC Phase 3) or worse, they are in need of urgent humanitarian food assistance. See description of IPC framework for classifying acute food insecurity here.
[3] According to BBC news reports, this is likely occurring as of mid-February 2022.

Summary of national context

Demographics and livelihoods of Ukrainians

Ukraine is a large country with strategic access to the Black Sea (Figure 1). According to the State Statistics Service of Ukraine, the country is home to a population of approximately 41.6 million people (2020 estimate), around 70 percent of whom live in urban areas. According to World Bank data, both urban and rural populations have been declining since the mid-1900s, due to emigration as well as high death rates and low birth rates. Life expectancy in 2019 was 66.9 and 77 for males and females, respectively, notably lower than in other European countries. As a result of these trends, Ukraine's population is aging. The national age dependency ratio (number of people younger than 15 or older than 64, as a percent of the working-age population (ages 15-64)) has been increasing since 2010, reaching 49 percent in 2020, with 17.4 percent of the population older than 64 (Figure 2).

Figure 2. Resident population by sex and age, as of January 1, 2021

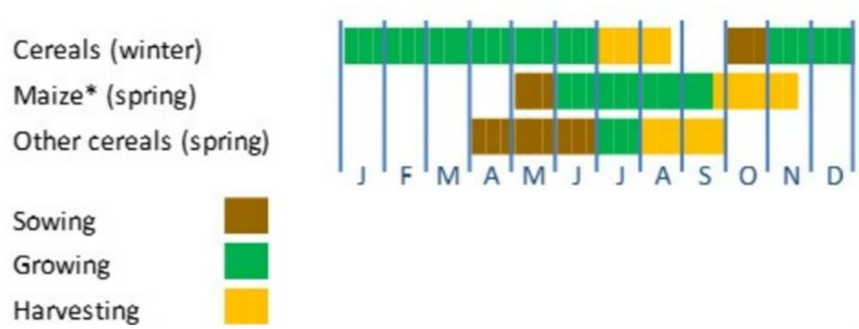


Source: *State Statistics Service of Ukraine*

Agriculture is important to Ukraine's economy and to rural livelihoods, contributing 11 percent of GDP in 2020. According to the State Statistics Service of Ukraine, agriculture provided 17 percent of the country's employment in 2020. According to FAO GIEWS, the agricultural season typically begins with planting of winter crops in the fall (Figure 3). Wheat is the most important winter crop, accounting for approximately 75 percent of total winter crop area in the 2020/21 production season,[4] according to USDA reporting. Following a dormancy period in the winter, crops grow in the spring. According to CHIRPS historical data, most precipitation is received from May to October, peaking in June and July. Harvesting of cereals typically occurs from July through November, beginning with winter cereals in July and August, followed by other cereals (mainly barley) in August and September, and concluding with maize in October and November (Figure 3). The long winter season—which typically lasts through around March—is traditionally a lean season when sources of food and income earning opportunities are limited for rural households, and when households are likely to have increased expenditure needs for heating.

In urban areas of the country, both formal and informal employment in labor commonly provides income for poorer households. In urban areas, of the employed urban population aged 15-70 years in 2019, "professionals" was the most commonly reported occupation group (22 percent) according to the State Statistics Service of Ukraine, followed by "services and sales workers" (17 percent) and "skilled workers using specific tools" (13 percent). "Plant and machine operators and assemblers" was reported by 11 percent, and "elementary occupations" by 9 percent of those surveyed, though these lower paying occupations are more important for poorer urban households. A total 14 percent of urban employment was in the informal sector in 2019, compared to 37 percent in rural areas. In Kyiv, aside from employment in government ministries, manufacturing and engineering industries provide considerable employment in factories, iron and steel plants, coal fields, and construction.

Figure 3. Cropping calendar in Ukraine



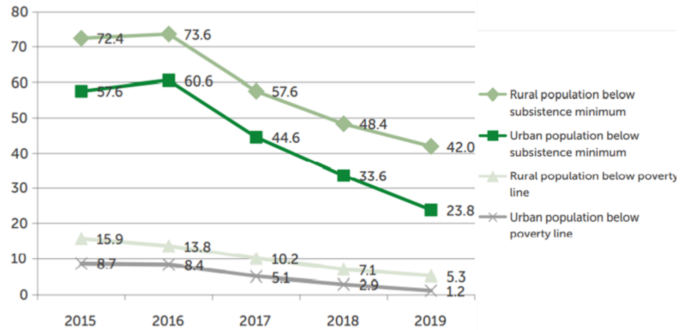
Source: *FAO GIEWS*

Ukraine has made progress reducing poverty in recent decades (Figure 4), though high unemployment remains a problem. According to the State Statistics Service of Ukraine, 79.2 percent of males and 68.9 percent of females of working age (15-59) participated in the labor force in 2019, and 72.2 percent and 63.2 percent, respectively, were employed. According to data from the State Statistics Service compiled

[4] A production season begins with planting and ends with harvesting, while a marketing year (MY) starts with harvesting and lasts through the rest of the year. For example, the 2020/21 production season is linked to MY 2021/22 by the harvest that occurred in 2021.

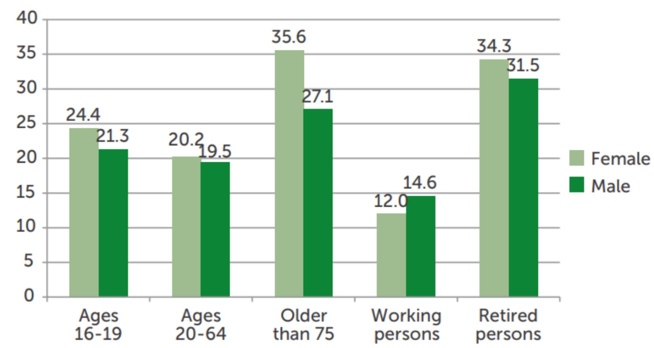
and analyzed by FAO, the population with income below the subsistence minimum—a measure calculated by the Ministry of Social Policy to reflect the changing cost of an expenditure basket of 296 items including the minimum essential food basket, non-food items, and essential services and used as a benchmark for social welfare but which has been criticized for being insufficient to meet real needs—declined from 2016 to 2019 in both urban and rural areas alongside reductions in poverty (Figure 4). However, among retired persons and among those older than 75, the share of the population that cannot meet minimum subsistence needs has remained high, at around one third (Figure 5). Additionally, households of older persons (60 and over) spend a greater share of total expenditures on food. While pensioners' households have been found to record sufficient food consumption (and better than other groups, contrary to popular perception) in terms of quantities consumed according to the Ukrainian Centre for Social Reforms, these households consume less diverse diets, poorer quality goods (such as processed rather than fresh meat products), and energy-dense but nutrient-poor food groups (such as fats), which have been linked to diet-related chronic diseases that are costly and detrimental to quality of life.

Figure 4. Percent of population with income below national poverty line (in grey) and below subsistence minimum (in green)



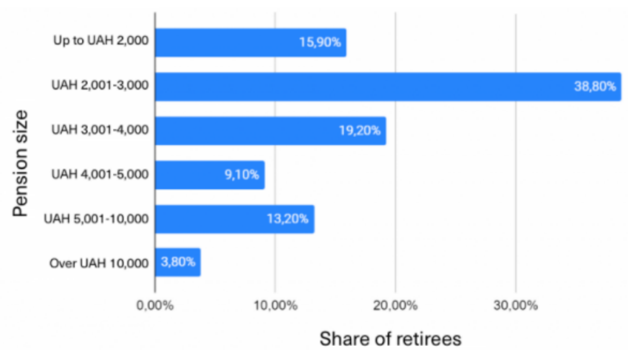
Source: FAO *using data from the State Statistics Service of Ukraine*

Figure 5. Percent of population with income below the subsistence minimum, by selected demographic groups



Source: FAO *using data from the State Statistics Service of Ukraine*

Figure 6. Pensions of Ukrainians as of April 1, 2021 (1 USD was equal to approximately 27 UAH in April 2021)



Source: *Vox Ukraine & Pension Fund of Ukraine*

According to data from the Pension Fund of Ukraine as reported by Ukrinform, the number of retirees in Ukraine stood at 10.8 million as of January 1, 2022. Despite declines in the number of retirees in recent years, Ukraine's pension system is considered unsustainable due to the increasing average age of the country's population. Ukrainian authorities have raised the retirement age for women several times since 2012 (when the retirement age for women was 55) and, in May 2021, the retirement age stood at 60 for both men and women. Ukraine has also been increasing the minimum years of required retirement contributions and plans to continue to do so until 2028; those who do not meet this requirement have their guaranteed minimum pension payment proportionally reduced. Overall, pension payment amounts are tiered, and most retirees receive the lowest pension amounts (Figure 6). Public servants—including judges, prosecutors, and educators—reportedly receive higher pension amounts than those in other occupation categories. Pensioners access payments through banks and paid postal facilities.

Agricultural production and exports

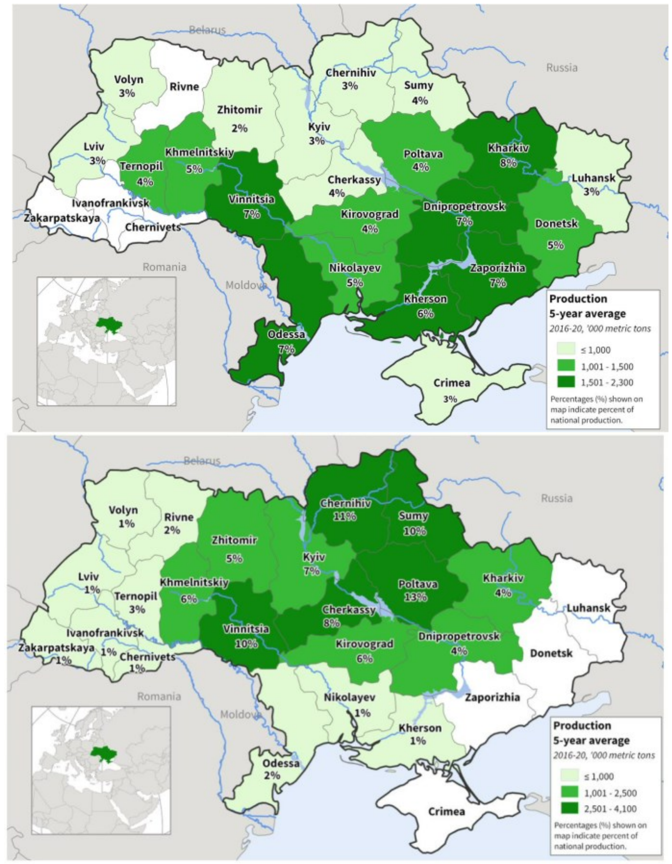
Ukraine is known for its fertile soil and has been historically considered a breadbasket of Europe. In the 2020/21 marketing year (MY) (linked to the 2019/20 production season by the 2020 harvest), the country's most important agricultural commodities in terms of production amounts were maize (30.3 million metric tons (MT)), wheat (25.4 million MT), potato (20.8 million MT), and sunflower seed oil (14.1 million MT). Barley and rye are other cereals produced in smaller amounts. Production of wheat has remained relatively stable since 1990, while production of maize has increased notably alongside its comparative profitability in global markets. Production of potatoes and sunflower seed have also increased. Of cereal production for domestic use in Ukraine, maize is predominantly consumed as animal feed whereas wheat is mainly used for human consumption and is therefore of greater importance for Ukraine's food security. The agricultural area east of the

Dnipro River typically produces around half of national wheat, maize, and sunflower seed production according to 2016-2020 averages from the State Statistics Service of Ukraine and presented by USDA (Figure 7). Large agricultural enterprises (of which there are around 45,000) produce an estimated 55 percent of Ukraine's gross agricultural output, with the remaining being produced by over 4 million households in smaller amounts. Agricultural enterprises generally specialize in the cultivation of export-oriented crops, while small-scale farmers produce for subsistence and internal consumption.

Ukraine has remained among the world's top ten exporters of wheat, maize, and sunflower oil for over a decade, due to agricultural production potential and favorable geographical position for trade. A gradually weakening domestic currency (against the USD) since 2014 has also supported strong exports. Grain exports are brisk in the first few months following the harvest, given limited storage capacity. Grain exports from Ukraine have increased markedly over the past decade, including in recent years (Figure 8). As a share of Ukraine's total production, 66 percent of wheat, 79 percent of maize, and 52 percent of barley was exported in MY 2020/21. Ukraine is also increasingly important for oilseeds and is the world's top exporter of sunflower oil, accounting for half of the world's sunflower oil exports in MY 2019/20. Sunflower seeds are mostly processed locally, increasing profit margins before oil is exported. In MY 2020/21, Ukraine was also the world's third largest exporter of rapeseed.

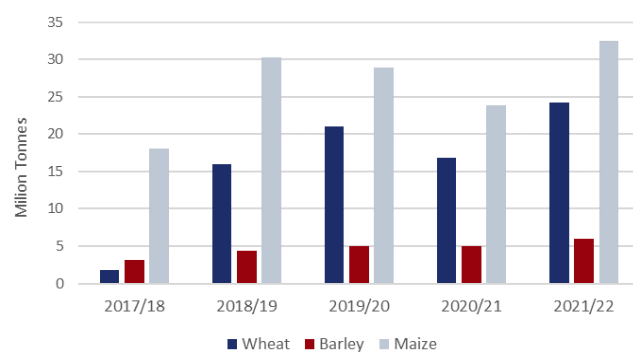
Ukraine's status as one of the world's major grain exporters makes the nation's food security less vulnerable to global supply and price shocks. In 2018, the country generated enough agricultural production to feed 2.6 times its own population. However, Ukraine's economy depends on revenue and foreign currency from agricultural exports (the largest export category since 2013), as do the livelihoods and typical income sources of millions of rural households, making the country's food security vulnerable to a shock to Ukrainian production or exports.

Figure 7: Ukrainian wheat production (top) and maize production (bottom), 2016-2020 average



Source: *USDA using data from the State Statistics Service of Ukraine*

Figure 8. Exports of Ukrainian wheat, maize, and barley, from MY 2017/18 to MY 2021/22 (MMT)



Source: *FEWS NET using USDA data*

In particular, Ukrainian wheat has become essential to the staple food supplies of developing regions, with much of Ukrainian wheat destined for Middle East and North African nations (Figure 9), many of which are dependent on imports. According to the most recent available FAO data, in calendar year 2020, Lebanon sourced 54 percent of its total wheat (grain and flour) imports from Ukraine, followed by Tunisia (49 percent), Libya (47 percent), Pakistan (43 percent), and Indonesia (27 percent). According to analysis carried out by IFPRI, using USDA Production, Supply, and Distribution (PSD) data and Trade Data Monitor (TDM) data, Tunisia, Egypt, Libya, Lebanon, and Pakistan sourced over 30 percent of their total wheat imports from Ukraine, on average from 2018 to 2020. In MY 2020/21, Africa, East and Southeast Asia, the Middle East, and South Asia were the largest regional destinations for Ukrainian wheat, importing 36 percent, 27 percent, 17 percent, and 16 percent, respectively, of Ukraine's total wheat exports that year. Also that year, Pakistan became a new market and the third largest importer of Ukrainian wheat, importing approximately 1.2

million MT. Additionally, Indonesia and Egypt began to source more wheat from Ukraine in 2021 following a doubling of Russian export taxes. The highest dependence on Ukrainian maize as a percent of total imports is recorded by China (at nearly 70 percent according to the same IFPRI methodology), the EU (nearly 60 percent), and Turkey (nearly 40 percent). Grain exportation occurs year-round according to IFPRI, with most wheat typically exported from July to November and most maize typically exported from November to June.

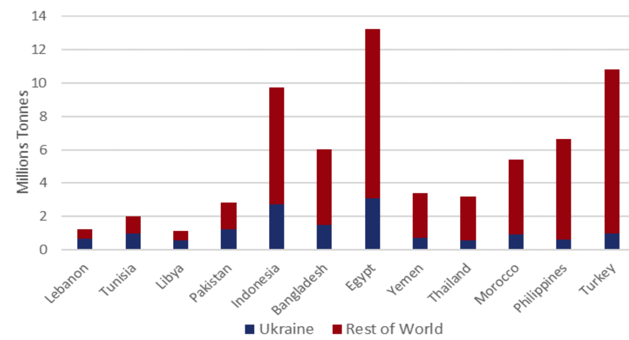
In the most recent 2020/21 production season (linked to the MY 2021/22 by the 2021 harvest), Ukraine is expected to have produced a record 81 million MT of cereals (Figure 11), 25 percent higher than last year and 21 percent higher than the 2016-20 average. Wheat production reached a record high of 33 million MT, about 30 percent above the previous year and 13 percent higher than MY 2019/20. A total 73 percent of domestically-produced wheat in MY 2021/22 was exported (amounting to roughly 24 million MT; the world's sixth largest exporter) compared to 66 percent in the previous year, attributed to greater exportable supplies. Domestic wheat consumption is forecasted to reach 27 percent of total production in MY 2021/22, compared to 34 percent in MY 2020/21, due to higher opening stocks in the beginning of MY 2021/22.

Wheat exports in MY 2021/22 are forecast at a record 24 million tons, 74 percent higher than the four-year average according to USDA data. According to USDA forecasts, in the ongoing MY 2021/22, Ukraine will be responsible for 12 percent of global wheat exports and 17 percent of global maize exports, as well as 18 percent of global barley exports and 15 percent of global rye exports.

Infrastructure and logistics

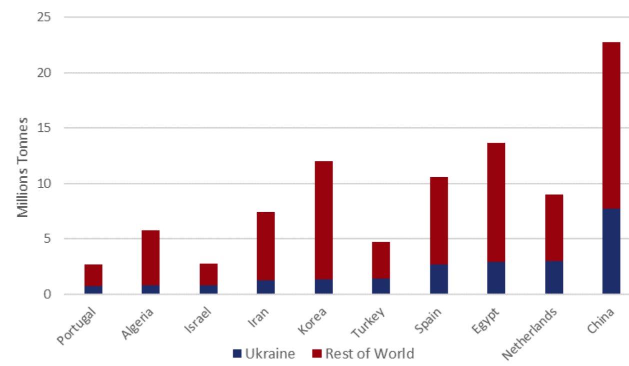
Local markets supply fresh food products in each region and city of Ukraine. However, most of Ukraine's domestic supply relies on large local and international players. The export of grain is in the hands of a few companies, most of which are large international agricultural commodity traders. These exporting companies often own grain elevators, private ports, rail wagons, and ships to transport, store, process, and export their own grain or grain purchased from other producers. Ukraine's existing storage infrastructure of 57 million tons allows simultaneous storage of 70-80 percent of the annual harvest of grains in 1,355 elevators (Figure 12). Poltava, Odessa, Vinnitsia, and Nikolayev have the most significant storage volumes in Ukraine, accounting for 34 percent of the total. The State Food and Grain Corporation has the greatest number of elevator facilities. The Ukrainian government holds strategic emergency reserves of different goods, including agricultural and food products. Purchases are made through open tenders. The State Reserve Agency of Ukraine has storage facilities across the country, including in Dnipro. Railroad and automobile roads provide 92 percent of the main transportation lines within Ukraine. Ukraine can ship over 40 million tons of grains and oilseeds annually through 13 seaports, predominantly on the coasts of the Black and Azov Seas (Figure 13). Over 60 percent of shipping volumes are concentrated in the Odessa region (Odessa, Illichivsk, and Yuzhny ports). In 2020, Chonomorsk and Pivdennyl (Odessa), and Mykolaiv (Nikolayev) ports held 35.5 percent of the total grain transshipment in Ukraine.

Figure 9. Sources of wheat and wheat flour of the top importers of Ukraine grains in calendar year 2020 (MMT)



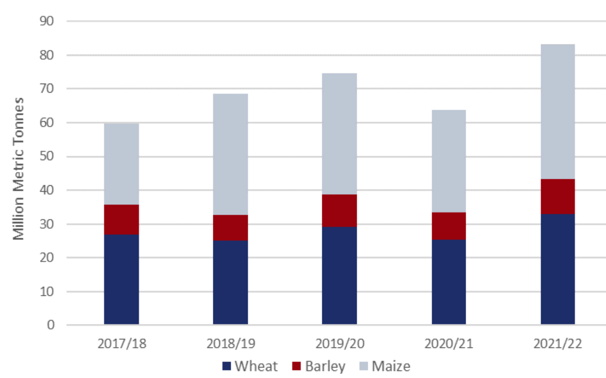
Source: *FEWS NET using FAOSTAT data*

Figure 10. Sources of maize and maize flour of the top importers of Ukraine grains in calendar year 2020 (MMT)



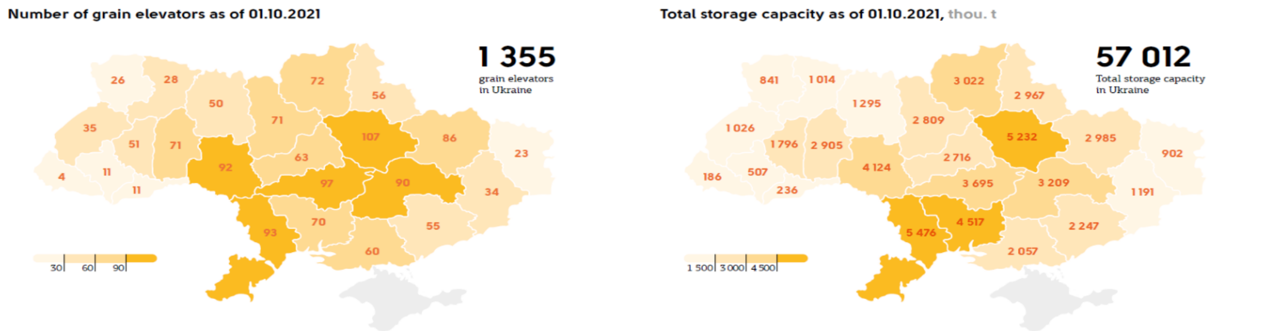
Source: *FEWS NET using FAOSTAT data*

Figure 11. Production of select grains in Ukraine from MY 2017/18 to MY 2021/22



Source: *FEWS NET using USDA data*

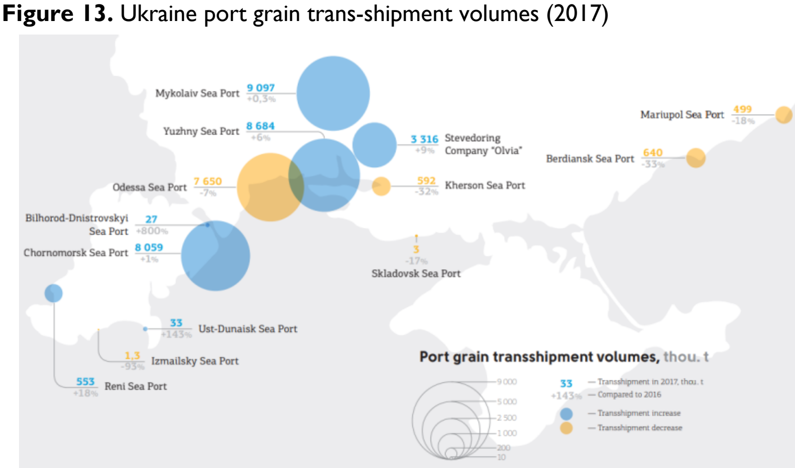
Figure 12: Ukraine: Grain elevators and storage capacity by region (2021)



Source: [Agribusiness Ukraine](#)

After the annexation of Crimea in 2014, average annual cargo volumes via Ukraine's Sea of Azov ports reduced dramatically. Export volumes through the port of Berdyansk have fallen tenfold, from 4.5 million tons annually prior to 2014 to [450,000 tons annually](#). In Mariupol, annual export volumes have approximately halved from 13 million tons in 2013 to a low of [5.8 million tons](#) in 2019. As such, while there exists the potential to significantly expand export volumes through the Sea of Azov ports, these ports face [challenges](#) with insufficient capacity of port railway stations and unsatisfactory railway connections.

Figure 13. Ukraine port grain trans-shipment volumes (2017)



Source: [Agribusiness Ukraine](#)

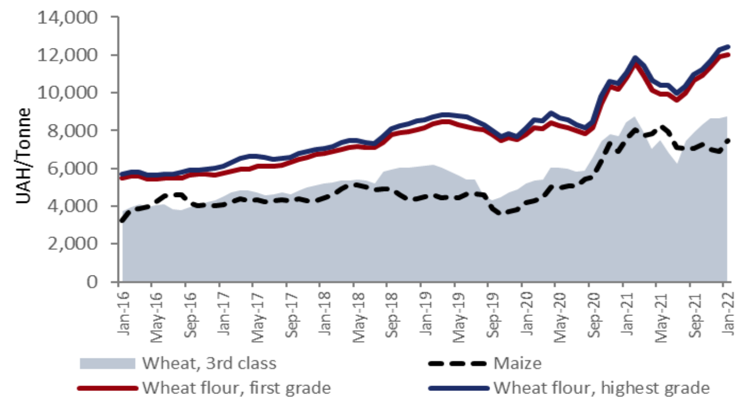
As a result, Ukraine's western Black Sea ports have been [increasing their capacity](#) in recent years to accommodate a growing volume of cargo shipments due to recent disruptions to trade links and resultant rerouting in the flow of goods, including when the Russians started to [actively impede navigation](#) on the Sea of Azov in 2018/2019. According to [USDA](#), current [storage capacity is expected to be limited](#) relative to current export volumes, resulting in pressure to export grains quickly.

Current food security situation in Ukraine

Though the economy and income earning have recovered somewhat in 2021, [supported](#) by above-average cereal production and exports as well as rising global prices, the COVID-19 pandemic has negatively affected livelihoods and income-earning for many of the poorest households in Ukraine, with the [informally employed](#) in urban areas among the worst affected. According to the National Bank of Ukraine, real GDP [contracted by 4 percent](#) in 2020, compared to 3.2 percent growth in 2019, driven by impacts of domestic control measures, global impacts on [trade](#) and supply chains, and reduced domestic demand and investment. Though rural livelihoods were more resilient, FAO assessments found that smallholder farmers, in particular, were negatively affected by control measures that impeded access to markets. In 2021, renewed control measures constrained economic GDP growth, reducing originally forecast 5 percent growth to under 3 percent growth in 2021, with per capita GDP in 2021 the lowest in Europe. However, given economic recovery and government-supported efforts to stimulate domestic demand, income-earning has improved for many Ukrainians. Government spending on wages, salaries, and pension payments have increased, though spending on residential services subsidies (including utilities) and rehabilitation for the disabled was [significantly reduced in 2020](#). [Wages and pensions rose 19 and 16 percent](#), respectively, in 2021, with [remittance inflows](#)—mainly from Russia— also high in 2021.

However, improvements in purchasing power have been significantly limited by rising prices of food and non-food commodities. Gas prices in the April-September 2021 period were double prices at the same time in 2020, increasing food processing and transportation costs. After increasing sharply between August 2020 and February 2021, domestic wholesale prices of wheat followed an overall declining trend until July 2021 but have been increasing since then (Figure 14). Meanwhile, domestic wholesale prices of maize peaked in May 2021 and declined slightly afterwards, but have overall remained at very high levels, mirroring trends in the export market. High prices of maize have also reportedly contributed to rising production costs in the livestock sector. At the end of 2021, annual inflation stood at a four-year high of over 10 percent. According to data from the State Statistics Service, annual food inflation in November 2021 stood at 13.9 percent. Bread prices had increased even more, by 18.1 percent. In early 2022, prices for grains and oilseeds are an estimated 20-30 percent above last year's levels, also putting upward pressure on prices of meat and dairy. In a December 2021 rating poll conducted by the Institute for the Future, 52 percent of surveyed Ukrainians reported that their financial situation had deteriorated, while only 10 percent reported improvement. No food price regulation has taken place since July 2017.

Figure 14. Historical wholesale commodity prices in Ukraine, January 2016 to January 2022



Source: *FEWS NET using FAO data*

Donbas

The Donbas region (Donetsk and Luhansk oblasts) in eastern Ukraine continues to be impacted by protracted conflict, now in its eighth year. According to the 2022 Humanitarian Needs Overview (HNO), about 38 percent of Donetsk and Luhansk oblasts remain outside government control, with the "line of contact" separating the government-controlled areas from the non-government-controlled areas. In addition to causing population displacement, conflict has eroded typical livelihoods and reduced income-earning opportunities, including by reducing access to agricultural land and restricting movement required for business and trade, and continues to impede provision of public services. At the national level, the unemployment rate among those 15 years and older stood at 8.2 percent in 2019. In the Donbas oblasts of Donetsk and Luhansk, however, this rate was 13.6 and 13.5 percent, respectively, the highest unemployment rates in the country. The population in this region is older than the national average because younger people have moved away. Approximately one-third of the population in conflict-affected areas are aged 60 and older, according to the 2022 HNO. According to FAO, pensions are the main source of income for most households in government-controlled areas of Donbas. Agriculture provides the main income source for an estimated 5 percent of households and is expected to provide an important source of food for many rural households, given resilient grain production levels and declining export opportunities in an environment of overall limited livelihood options. In non-government-controlled areas, many people (especially elderly pensioners) ordinarily regularly cross the contact line in order to receive social benefit payments including pensions. However, COVID-19 restrictions have significantly reduced the number of recorded crossing and, while the government responded with policy measures to facilitate continued access to pensions and other social benefits, it is likely that access to these benefits has reduced. Although some humanitarian organizations including UNCHR provide support to vulnerable people in non-government-controlled areas, overall few partners have been able to implement cash transfer programs in these areas. It should be noted, however, that data collected by the cash working group which surveyed vulnerable households mostly within 5km of the line of contact between July 2020 and June 2021 showed that households of elderly (60 and older) earned the highest average monthly income per person of any vulnerable demographic group analyzed. Households with unemployed people aged 50-59 and households with single parents earned the lowest.

According to FAO, Luhansk and Donetsk reported a sharp decline in agricultural production after 2013, resulting from mobility constraints due to conflict. Production of fruits, vegetables, and dairy was significantly affected due to Ukrainian trade restrictions with Russia and non-government-controlled areas of Donbas. However, grain production was more resilient and did not significantly decline. According to USDA, the Donbas region produces 8 percent of national wheat production (3 percent in Luhansk and 5 percent in Donetsk) and 9 percent of national sunflower seed production (5 percent in Luhansk and 4 percent in Donetsk). The Donbas region was also responsible for 5 percent of national barley production.

Linkages with the export economy were affected by limited access to industrial facilities and trade restrictions, increasing the relevance of these regions for the local market. Luhansk and Donetsk are relatively detached from the main Black Sea seaports of Odessa, Ilyichevsk, and Yuzhniy; however, Donetsk oblast lies along the northern coast of Azov Sea where the Ukrainian merchant sea ports of Berdyansk and Mariupol enable the export of staple grains and other commodities via the Sea of Azov. Overall, persistent conflict since 2014 continues to disrupt grain logistics and infrastructure in Donetsk and Luhansk. As a result, grain and oilseed transportation and exports are irregular, depending on the intensity of the conflict and insecurities, according to FAO.

Current food security outcomes

Despite some economic recovery in 2021, poor households in urban areas of Ukraine are likely still struggling with the prolonged impacts of COVID-19-related disruptions to income-earning. Coupled with rising food prices in 2021, some worst-affected urban households are likely already facing Crisis (IPC Phase 3) outcomes characterized by food consumption gaps or engagement in damaging livelihood coping strategies. Across the country, an estimated 0.5-1.0 million people are expected to require humanitarian assistance, with the highest concentration among poor households in urban areas and in the Donbas region.

In rural areas, winter crops are currently in dormancy stages and the typical lean period is ongoing. However, given the bumper harvest in the recently concluded 2020/21 production season as well as increased spending on pensions and social safety nets, many poor rural households likely have above-average levels of food stocks and/or monetary resources. As such, most rural households are expected to be meeting their food and essential non-food needs, though high prices of food and non-food commodities are likely constraining resources for households who harvested smaller amounts or who are otherwise relatively more dependent on markets for food and/or on fuel for transportation and livelihoods. Among the worst affected are likely poor households with smaller landholdings or without a primary income-earner, including many of the disabled, widowed, and elderly, among whom Stressed (IPC Phase 2) outcomes are likely.

Typically, rural Ukrainian households consume less diverse diets than urban households, relying more on grains, oils, starchy vegetables, and sugars. Dietary diversity is currently expected to be at seasonally low levels. Though above-average resources from the harvest and government support would be expected to support better-than-normal dietary diversity, this has likely been limited by rising food prices, with overall levels of dietary diversity expected to be fairly typical. Less than ten percent of rural households have access to centralized sewer networks according to FAO, with poor sanitation likely to be inhibiting proper nutrient utilization among worst-affected households.

In Donbas, according to a non-representative food security and livelihoods assessment of mostly female-headed households and pensioners conducted in May 2020 to understand the impacts of COVID-19, rising costs of food and essential non-food commodities was the primary negative impact experienced by most respondents, though around one-third also reported a decrease in income. Stressed (IPC Phase 2) level livelihood coping strategies were commonly reported, including reducing essential expenditures (66 percent of respondents) and spending savings (34 percent). Additionally, 39 percent reported "poor" food consumption. Given these results and the characteristics of the sample, Stressed (IPC Phase 2) outcomes are also expected among the general rural population, though a relatively higher share is likely facing worse outcomes than in other rural areas of Ukraine.

Scenario of "moderate to major" invasion by Russian forces

Scenario parameters

Among the possible conflict scenarios, this analysis assumes that Russia attempts the Dnipro River strategy (Figure 15) in a military effort that lasts for approximately the next three months, and invades Ukraine on the southern and eastern front, while threatening the capital but not yet waging a full invasion towards Kyiv (Figure 16). This would entail Russia advancing along two routes: the central route and the southern route. Through the central route, Russian forces would advance in three directions: from Belgorod Russia to Poltava Ukraine, through Kharkiv; from Donetsk to Zaporizhzhia and Dnipropetrovsk; and along the southeastern coast from Rostov-on-Don, Russia through Mariupol, Berdyansk, and Melitopol, Ukraine. Through the southern route, Russia would advance from Perekop Isthmus to Kherson and Melitopol. This scenario assumes it is unlikely Russia advances towards Kyiv in the near term (during the next three months).

Invasion of Ukraine through these routes would directly impact Luhansk, Donetsk, Kharkiv, Poltava, Dnipropetrovsk, Zaporizhzhia, and Kherson oblasts. The total population of these oblasts was estimated in 2020 at 13.4 million, or roughly 32 percent of the population. Within these oblasts, the most densely populated cities include Kharkiv (around 1.5 million people) and Dnipro (around 1 million people). Similar to the national trend, the population in these oblasts is increasingly urban and elderly. Roughly 25 percent of the population in these areas is over the age of 60, and an estimated 78 percent lives in urban areas.

Though not part of the scenario analyzed, it is possible that Russia would additionally engage in a full-scale naval blockade of the Black Sea, where it has been pursuing increased control over the years. Should this occur, the domestic and international impacts resulting from reduced Ukrainian exports outlined below would be significantly exacerbated.

Domestic impacts on livelihoods and production

In the short term, significant disruption to livelihoods—including during the agricultural growing season—would be expected in affected areas east of the Dnipro River (both rural and urban) due to the direct impacts of conflict on the ground along invasion routes and from airstrikes targeting critical infrastructure including communication towers. In addition to constraining access to agricultural land and other livelihood and income-earning activities, conflict would likely result in significant damage to homes, productive assets, and agricultural land (particularly if Russian forces engage in scorched earth tactics), as well as to roads and other civilian infrastructure including bridges (including many in urban areas). As territory changes hands, millions of people would be at risk of losing access to critical income sources, including government-provided pensions and social safety nets. Active conflict would also result in high levels of population displacement as households in affected areas east of the Dnipro River flee to safer areas, especially to western Ukraine, but also with some fleeing farther west into Poland, elsewhere in Europe, and ultimately to other neighboring countries including Russia. Hundreds of thousands to millions of displaced households separated from assets and typical sources of income would likely need immediate assistance, particularly during the ongoing winter season when fuel needs are highest. Households without sufficient resources living in temporary locations would be expected to need sustained humanitarian assistance until livelihoods are re-built in new, permanent locations. Growing displaced populations would also place additional pressure on community resources in western Ukraine. Other populations would be unable to flee due to lack of resources or significant challenges in crossing lines of control. Access constraints will likely prevent humanitarian actors from providing assistance to some populations that remain in place. Displacement to areas outside of Ukraine would occur in larger numbers if the scale of invasion and displacement exceeds what is anticipated in the scenario analyzed.

Figure 15. Several plausible conflict scenarios

Withdrawal: Russian redeploys some of its ground forces away from the Ukrainian border; negotiations are successful, but Russia continues to aid pro-Russian rebels in eastern Ukraine
Donbas Invasion: Russia sends conventional Russian troops into the breakaway regions of Donbas and refuses to withdraw them until peace talks end and Kyiv agrees to implement the Minsk Accords.
Dnipro River Strategy: Russia attempts to seize territory as far west as the Dnipro River to use as a bargaining chip or to incorporate this territory into the Russian Federation
Black Sea Exclusion: Russia attempts to seize Ukrainian territory up to the Dnipro River and an additional belt of land (including Odessa) that connects Russia with the Transdniestria Republic and separates Ukraine from access to the Black Sea.
Transdniestria Option: Russia attempts to seize only the belt of land between Russia and Transdniestria – including Mariupol, Kherson, and Odessa – to secure freshwater supplies for Crimea and block Ukraine's access to the seas, while avoiding major combat over Kyiv and Kharkiv.
Full invasion: Seize all of Ukraine and, with Belarus, announce the formation of a new tripartite Slavic Union.

Source: *Russia's Possible Invasion of Ukraine, CSIS*

Figure 16. Conflict scenario of this analysis under the Dnipro River Strategy scenario (shown in the blue box of Figure 15)

Source: *Russia's Possible Invasion of Ukraine, CSIS*

Active conflict would likely disrupt agricultural activities during a critical growing period and would likely damage crops once they reach vegetative stages in early spring. The greatest impacts would likely be felt in southeastern oblasts along invasion routes, including in Zaporizhzhia, Kherson, Dnipropetrovsk, Kharkiv, and Poltava, which produce a variety of cereals (Figure 17). Overall, Ukraine's 2022 production of cereals and other key crops including oilseeds could be significantly reduced (depending on the scale of crop damage and reduction in yields) due to the high concentration of favorable agricultural land and production east of the Dnipro River—an area responsible for around half of national wheat, maize, and sunflower seed production. Additionally, Ukraine is dependent on imports for much of its agricultural input needs. Impacts of conflict and further geopolitical instability could negatively affect farmers' ability to access sufficient or high-quality inputs—particularly among farmers in the east—including seeds and fertilizer. This could reduce yields and reduce or delay planting of spring crops, which typically occurs from April to June. Rising prices of fuel would further constrain agricultural households' ability to afford agricultural inputs. With slightly above-average area planted in the ongoing 2021/22 production season and average precipitation currently forecast[5] according to most international ensemble forecast systems, another favorable production season is currently anticipated in western Ukraine, though disruptions to supply chains which might impede access to key inputs, perhaps significantly, and lower precipitation amounts could dampen agricultural production prospects.

In conflict-affected areas and in areas impacted by supply chain disruptions due to infrastructure damage along key trade routes, market supply and access for both traders and households would likely be disrupted for variable periods of time. Periods of localized supply shortages would be possible, particularly for imported goods but also for some domestically produced commodities in areas where stocks are at low levels due to the current point in the season (as the next harvest will not start until around June). It would be possible that Russian forces cut off supply of food and essential non-food commodities—such as fuel and medicine—from particular areas as a strategy to pressure Ukrainian forces and authorities. Food prices would likely increase where supply is disrupted. Most major markets would be at risk of supply chain disruptions and, as such, while some disruptions to local market operations are also anticipated due to active conflict in the vicinity, supply chain disruptions pose widespread risks to market functioning throughout eastern Ukraine.

In addition to impacts on production and local supply, eastern Ukraine's capacity to store, transport, and export staple grains would be reduced by damage to—and changing control over—infrastructure. According to recent estimates, around 40 percent of all grain elevators and storage capacity are found east of the Dnipro River. Even more significantly, access to Black Sea ports critical for exports would likely be cut off, and international sanctions (on Russia) could prevent exports from non-government-controlled territory, significantly reducing export levels from eastern Ukraine. This would have a massive negative economic impact on eastern Ukraine, but also on the national economy. Due to the importance of agricultural exports for government revenue and foreign exchange, Ukraine would likely experience economic contraction in 2022. While most severe losses in economic activity would occur in eastern occupied territories, conflict would also likely sever existing value chains in integrated markets, potentially leading to reduced production and/or higher production costs in non-occupied areas. Although some producers of exportable agricultural commodities in western Ukraine could benefit from rising global prices and increased demand, these benefits would be limited by the capacity of storage and port infrastructure and the real possibility of canceled orders by importing countries.

Other livelihood and business activities in eastern Ukraine would be disrupted and damaged by conflict in the near term and would continue to be eroded in the medium and long term. Of note is Barabashovo market in Kharkiv, due to its relative economic importance and its reliance on trade with Russia and the neighboring Donbas region. Barabashovo—a commercial hub rather than a food market—is the largest market in Ukraine and one of the largest in Central Europe. Overall, conflict would damage the national and local economies, with levels of unemployment likely to increase. Areas west of the river would also be negatively impacted by economic consequences of an invasion. Russia's long-term objective could be to render Ukraine an economically inviable state.

The conflict since 2014 across Donbas (Luhansk and Donetsk oblasts) reduced access to the rest of the country. While the importance of the region as a supplier to the rest of Ukraine and to the global export market has since diminished, the importance in supplying local populations has increased. Territory east of the Dnipro River would likely experience similar shifts. Though increased local supply due to reduced exports would likely increase availability of food for farmer's own consumption and put downward pressure on prices of locally produced commodities, rising production costs are likely to place opposing upward pressure on prices of locally produced commodities, and prices of imported goods would also likely increase. Overall, reductions in income-earning could outweigh any benefits of increased food available for farmers' own consumption. In the near term, local agricultural production would also be threatened by the potential for shortages of inputs as well as any direct security threats at critical times in the current growing season.

5 Note: One ensemble forecast system, the North American Multi-Model Ensemble, recently revised forecasts to use February 2022 initial conditions, and the likelihood of below-average precipitation has increased, with average to below-average precipitation now forecast across the country. Below-average precipitation is a possible outcome that would further reduce production prospects. Other ensemble forecast systems consulted include the World Meteorological Organization and the Copernicus Climate Change Service, with January 2022 initial conditions.

Projected food security outcomes

Given the bumper harvest in the recently concluded 2020/21 production season, many poor rural households likely have above-average levels of food stocks and/or monetary resources. This is likely to provide some ability to cope with temporary disruptions to income-earning. However, resources among poor and smallholder households in rural areas east of the Dnipro River would be stretched thin as time goes on in the cold winter and lean months, with Stressed (IPC Phase 2) outcomes likely to persist until food and income associated with the next harvest become available around June/July. During this time, worst-affected rural households without sufficient resources would likely experience Crisis (IPC Phase 3) outcomes in the short term as stocks are exhausted during the winter months and livelihoods significantly disrupted. However, households whose food and/or income from agricultural production are significantly reduced by the impacts of conflict would not experience normal seasonal improvements.

Urban areas—being strategic and highly populated—would be devastated by the direct impacts of active conflict. Civilian casualties, high levels of displacement, and significant disruptions to livelihoods, income-earning, and markets would be likely. As soon as any invasion begins—or as the threat of invasion becomes more imminent—urban households who can afford to do so would likely stockpile food, driving up prices.[6] Poor households who cannot afford to stockpile food would likely face consumption gaps and Crisis (IPC Phase 3) outcomes during any periods of time where active conflict prevents households from accessing markets for several days or more.

In both rural and urban areas, poor households without a primary income-earner, including many of the disabled, widowed, and elderly, are likely to be worst affected by losses of typical food and income sources. While an increase in remittances would be expected as the diaspora increases support to family and friends in Ukraine, it is likely that financial transfers and services would be disrupted immediately as territory changes hands, preventing households from accessing remittances, pension payments, or other resources held in banks and, in the medium and long term, international sanctions on Russia could prevent households in occupied territory from receiving remittances until the international community takes steps (including by issuing waivers) to grant the needed permissions. Sanctions would present similar challenges to humanitarian actors seeking to assist Ukrainians in occupied territory, which would hinder the response and disrupt key sources of income for households currently receiving assistance (especially in Donbas). In such a context, should food supply be cut off from particular areas, Emergency (IPC Phase 4) outcomes might occur and would last until supply is restored.

Though significant uncertainty exists, an estimated 2.5-4.99 million people (around 5 to 10 percent of the national population)—mostly displaced households—would likely need humanitarian assistance in the event of an invasion of this scale.

International supply and market outlook

In the event of invasion, a slight increase in exports from western production areas—and from any eastern production areas that can export via alternative ports—might be expected due to high international demand and reduced exports from the east. However, ability to expand export volumes would likely be limited due to the country's limited storage capacity relative to current production and export levels. It is also likely that the impacts of international sanctions would pose a significant barrier to the expansion of exports from Russian occupied territory in eastern Ukraine. Overall, exports of key agricultural commodities—including wheat, maize, and plant oils—from eastern Ukraine would likely notably decline in 2022 under this scenario. Should Russia engage in a naval blockade of the Black Sea, an even more precipitous drop in exports would occur.

World Bank commodity price forecasts published in October 2021 indicated that global grain and oil prices (as measured by aggregated price indices) would stabilize in 2022, though at already high levels after increasing by 22 percent for the "grains" index and 40 percent for the "oils and meals" index in 2021. Wheat prices were projected to remain broadly stable in 2022, while maize prices were expected to decline slightly. However, the World Bank noted that the forecast was subject to risks including volatile input prices (especially for energy and fertilizer), biofuel policies (should production be diverted for biofuel), dry conditions due to the forecast La Niña (which has since emerged), and macroeconomic uncertainties. Given the current tensions in Ukraine, upward pressure on commodity prices (including energy, as Ukraine is a key supplier of Russian oil and gas to Europe), beyond what was forecast, is already occurring and will likely continue—regardless of whether an invasion occurs—due to the existing uncertainty for global markets. Beyond this, should Russia invade eastern Ukraine, the likely sharp reduction to Ukrainian exports—and any impacts of international sanctions on Russia that would disrupt Russian energy and grain exports, including Russian counter-sanctions and export restrictions—would significantly further disrupt international

[6] According to BBC news reports, this is now likely occurring as of mid-February 2022.

grain markets due to implications for energy markets and the high share of Ukrainian and Russian grains in global cereal exports. Given generally favorable global production forecasts, many buyers would likely source wheat and maize, or cheaper substitutes, from alternative markets. However, given the high share of Ukrainian grains in total global exports, a significant shock to Ukrainian grain exports would reduce supply relative to earlier forecasts, driving up prices of key food commodities including wheat, maize, and edible oils. It additionally remains possible that global production forecasts continue to be revised downward, including due to drought conditions related to La Niña and rising fertilizer prices. FAO is expected to publish its first production forecasts for 2022 harvests in the coming weeks.

Should this notable shock to exports occur, many importers of wheat—an important crop for food security—would pay higher international prices for wheat sourced from the international market. This would most significantly impact poor countries who are highly reliant on imports for their staple food supplies, as has been seen in previous global food price crises. Rising prices of edible oil—also imported by many poor countries and a staple of poor households' diets—would exacerbate detrimental food security impacts. Higher staple food prices in import-dependent countries would further constrain resources of poor households, exacerbating the scale and severity of existing food insecurity and increasing assistance needs, with poor urban households among the worst affected due to high market dependence. Additionally, the consequent increased dependence on humanitarian assistance would harm local livelihoods and economies and increase vulnerability to future shocks for millions of people in the world's poorest and already food insecure areas.

Several countries in the Middle East, North Africa, and Asia regions (including Lebanon, Tunisia, Libya, Pakistan, and Indonesia) have relatively higher dependence on Ukrainian wheat and would need to source grain from other countries at higher prices. The degree of concern for their ability to do this depends on a complex set of factors beyond their dependence on Ukrainian grain, overall import dependence, and overall financial status, including: the structure of importation supply chains including the number and nimbleness of major traders (many of whom are private companies), the country's geographic location including proximity to alternative source markets, the amount of staple food commodities grown domestically, existing trade policies, the strength of national institutions and ability of the government to respond quickly with appropriate trade and social safety net policies, the degree of international and regional cooperation in response, and more. Since the 2008 global food price crisis, there has been an increase in research seeking to understand and quantify vulnerability to trade shocks. However, efforts to develop trade vulnerability indices remain in early stages and disagreement exists over the utility of such efforts given complexities at both conceptual and empirical levels.